LESSON PLAN

GAS TURBINE ENGINE (16)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to NDT, need advantages, various types of defects in the material types of defects detected by this methods, Types of NDT metho
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to NDT, need advantages, various types of defects in the material types of defects detected by this methods, Types of NDT methods
5. Specific Objectives	16.1.1.1 Trainee will be able to describe types of NDT and perform liquid penetrant, magnetic particles and eddy current inspections

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of NDT types and defect classifications using L & Ds and IB; Instructor supervision of theoretical understanding.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Non-Destructive Testing?
(b) Why is NDT essential for Gas Turbine Engine maintenance?
(c) List three common defects found in engine materials.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and scope of NDT • Advantages of NDT over destructive testing • Classification of material defects • Overview of NDT method types	L & Ds	PPT, IB	15 min
(b)	• Internal vs external defects • Cracks, porosity, and inclusions • Corrosion and fatigue defects • Detection capabilities of NDT	L & Ds	PPT, PDF	20 min
(c)	• Liquid Penetrant method basics • Magnetic Particle method basics • Eddy Current method basics • Selection criteria for NDT methods	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the advantage of NDT regarding component integrity.
(b) Differentiate between surface and subsurface defects.
(c) Identify which NDT method is suitable for non-ferrous materials.

12. Home Assignment and Follow Up	Prepare a brief note on the limitations of Liquid Penetrant testing to be discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Liquid penetrant, limitations, physical principles, description of process, penetrant system liquid penetrant materials, types of developer, technique using portable equipment	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min